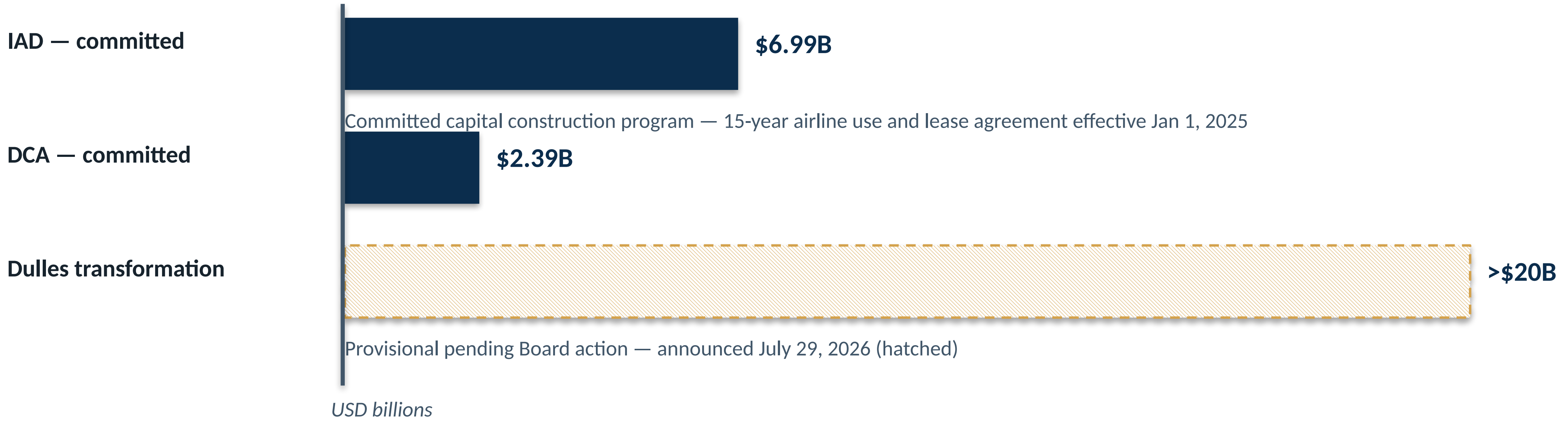
The loud terminal is the unintelligible one

A communications-integrity standard for MWAA terminals — argued as safety engineering, delivered through the Design Manual.

Technical read-ahead · Metropolitan Washington Airports Authority · August 2026

Sources: Transform Airports AI Council technical read-ahead, August 2026. Companion to the verified report ‘The Loud Terminal Is the Unintelligible One.’

Roughly $9B committed, $20B+ provisional — all of it passes through the Design Manual



Every square foot of that work passes through the MWAA Design Manual — “a mandatory guide with the force of law on the airport property.”

Sources: MWAA use-and-lease press release (2024); July 29, 2026 joint announcement as reported by Travel Agent Central and Simple Flying ($20B+ provisional pending Board action); MWAA Design Manual description, mwaa.com.

After January 29, 2025, no board signs a policy paraphrasable as 'fewer safety broadcasts'

THE CONSTRAINT

The NTSB's final report on the January 29, 2025 midair collision over the Potomac — 67 fatalities — resets public tolerance for any policy paraphrasable as "reduce safety announcements."

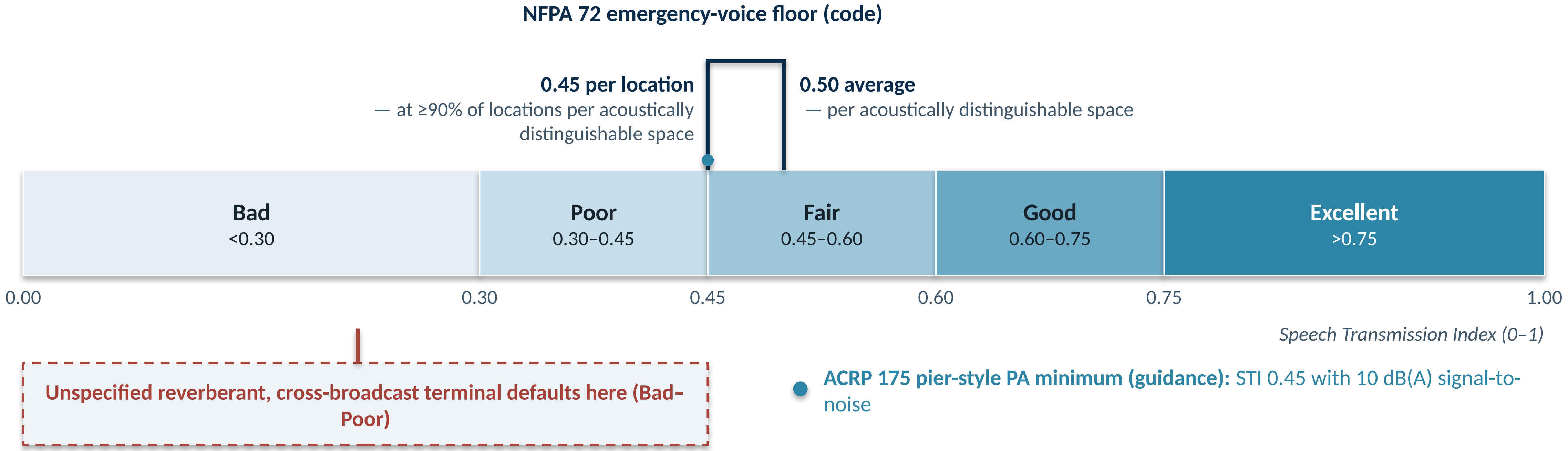
THE INVERSION

The standard is defensible only as the opposite of subtraction: a floor that raises the probability the announcement that matters — the gate change, the shelter-in-place, the IROPS diversion — reaches the passenger who needs it.

Sources: NTSB Aircraft Accident Report AIR-26/02 (final, 2026, ntsb.gov); contemporaneous coverage (CNN).

One ruler decides the argument: the Speech Transmission Index

A terminal fails these floors by being loud, reverberant, and cross-broadcast — precisely the unspecified default.



Code: NFPA 72 (adopted edition governs) · Guidance: ACRP Research Report 175 (TRB, 2017) · Measurement method: IEC 60268-16:2020

Sources: NFPA 72 (code paywalled; thresholds corroborated across 10+ records reproducing code language); ACRP Research Report 175 (TRB, 2017); IEC 60268-16:2020 (standard paywalled; bands corroborated in secondary records).

The Joint Commission did not silence ICU alarms — it made hospitals govern them

The problem, documented

One academic medical center logged 74,535 alarms across eight units in one week (illustrative magnitude). Clinicians had stopped hearing them.

Jan 1, 2014 — NPSG.06.01.01 phase one

Every accredited hospital must inventory and prioritize its clinical alarms.

Jan 1, 2016 — phase two

Governance policies, accountability, and proof on the record — not silence.

The pattern maps directly to airports: inventory, rank, govern, log.

Sources: The Joint Commission, National Patient Safety Goal NPSG.06.01.01 (phased 2014/2016); Respiratory Therapy Magazine alarm-fatigue review (per-bed figures vary by study; magnitude illustrative).

The 'silent airport' peer set is a category error — none published measurement

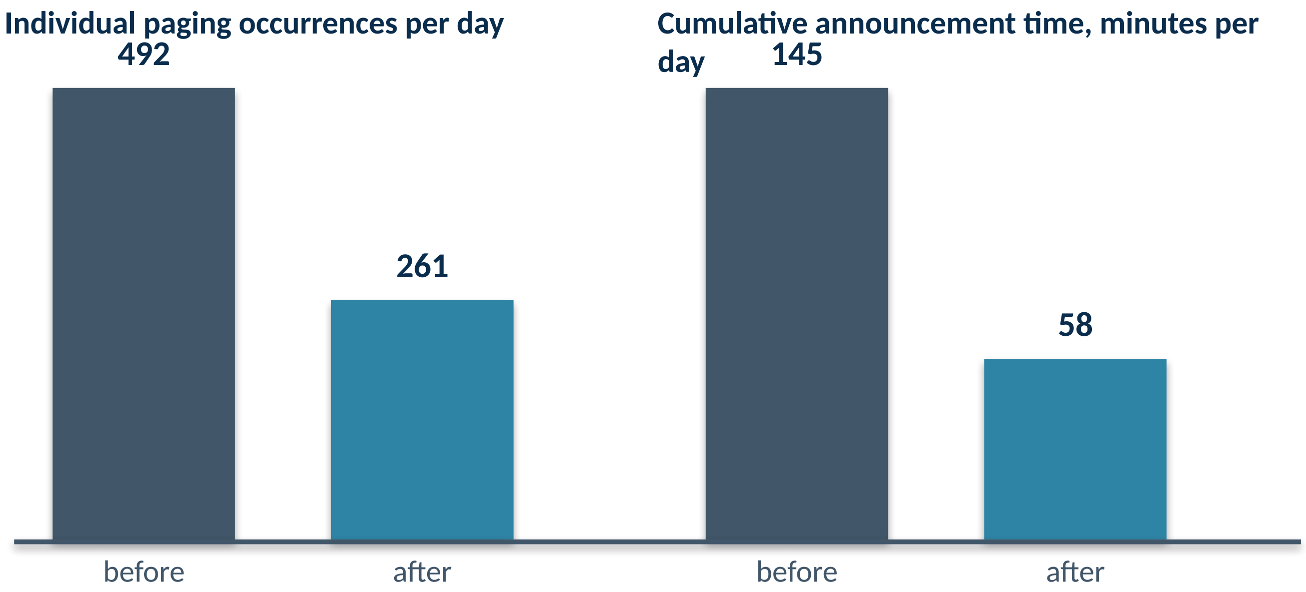
Airport	Announcement policy (operator-described)	Structural profile vs. IAD/DCA	Published intelligibility or missed-boarding data
London City (LCY)	Silent policy since 2008; no flight or gate announcements since Aug 18, 2016, except essential and emergency	Small, point-to-point, business-traveler-dominant	**None identified in this research pass**
Helsinki-Vantaa (HEL)	Silent regime since 2015; gate-area and emergency overrides retained	Small transfer airport, high English proficiency	
Copenhagen (CPH)	Gate and boarding information moved to screens and app	Point-to-point-weighted Northern European hub	
Amsterdam Schiphol (AMS)	Operator-described “Silent Airport”; information on screens and airport app	Large transfer hub — the closest structural analogue; rests on screen density and app-carrying passengers	

Policy analogues, not design precedents.

Sources: operator self-descriptions — londoncityairport.com; Finavia (Helsinki-Vantaa); Copenhagen Airport passenger-communication policy; Amsterdam Schiphol Silent Airport pages. No published intelligibility or missed-boarding measurement identified for any of the four.

SFO shows the operating model — and what subtraction without engineering costs

SFO-reported; methodology not published



THE SAME RECORD

Documented passenger complaints and missed-flight accounts attributed to the reduced-announcement practice, in contemporaneous reporting.

Announcement reduction without intelligibility engineering and channel redundancy is subtraction, not design.

Sources: SFO Quiet Airport outcomes as reported by International Airport Review (2023) and View from the Wing, citing SFO; all figures SFO-reported, methodology not published. Counterweight: Simple Flying, documented missed-flight and confusion accounts.

The people who most need to hear are the legal floor, not an edge case

28 CFR 35.160 — ADA Title II Effective communication with people with disabilities; auxiliary aids and services.

2010 ADA Standards §219 / §706 Assistive-listening systems in each assembly area where audible communication is integral to use.

14 CFR Part 382 Prompt access to the same gate information for passengers needing visual or hearing assistance.

guidance

ACRP Research Report 239 Digital-first substitution is partial, not sufficient, for travelers with disabilities and older adults.

A standard that mandates redundant visual, digital, staffed, and hearing-loop channels binds MWAA harder than current practice, not softer.

Reducing broadcast without those redundancies compounds exclusion.

Sources: 28 C.F.R. § 35.160 (ada.gov); DOJ 2010 ADA Standards §§219, 706; 14 C.F.R. Part 382 (ecfr.gov); ACRP Research Report 239 (TRB, 2023).

A two-tier standard — and the paragraph the silent-airport literature omits

MANDATORY AT BOTH AIRPORTS

- Acoustically-distinguishable-space (ADS) map as a design deliverable
- Intelligibility floors per NFPA 72 and ACRP 175
- Measurement per IEC 60268-16:2020, commissioned under representative traffic, with cure period
- Accessibility floors: 28 CFR 35.160, 2010 ADA Standards, 14 CFR Part 382
- Hearing conservation per 29 CFR 1910.95
- Audit-logged announcement governance
- Concessionaire TVs and gate-lounge audio kept in scope

CONTEXT-DEPENDENT BY TERMINAL

- Tier levels for PA zoning and announcement density
- Quiet and low-stimulation space scope
- Enforcement mechanics by lease type

IROPS OVERRIDE — during declared irregular operations, the Airport Duty Manager holds named authority to exceed announcement-density templates; every deviation is logged and reconciled afterward.

Sources: framework per the verified report; floors per NFPA 72, ACRP 175, IEC 60268-16:2020, 28 CFR 35.160, 2010 ADA Standards, 14 CFR Part 382, 29 CFR 1910.95; governance capabilities per ACI-NA (Feb 27, 2026).

The ask: write the paragraph this capital cycle

1
Office of Engineering drafts the amendment

2
Primary-text verification: Design Manual acoustic content; AUL majority-in-interest

3
Airline and tenant consultation; formal comment window

4
Board information item — resolution only on material objection

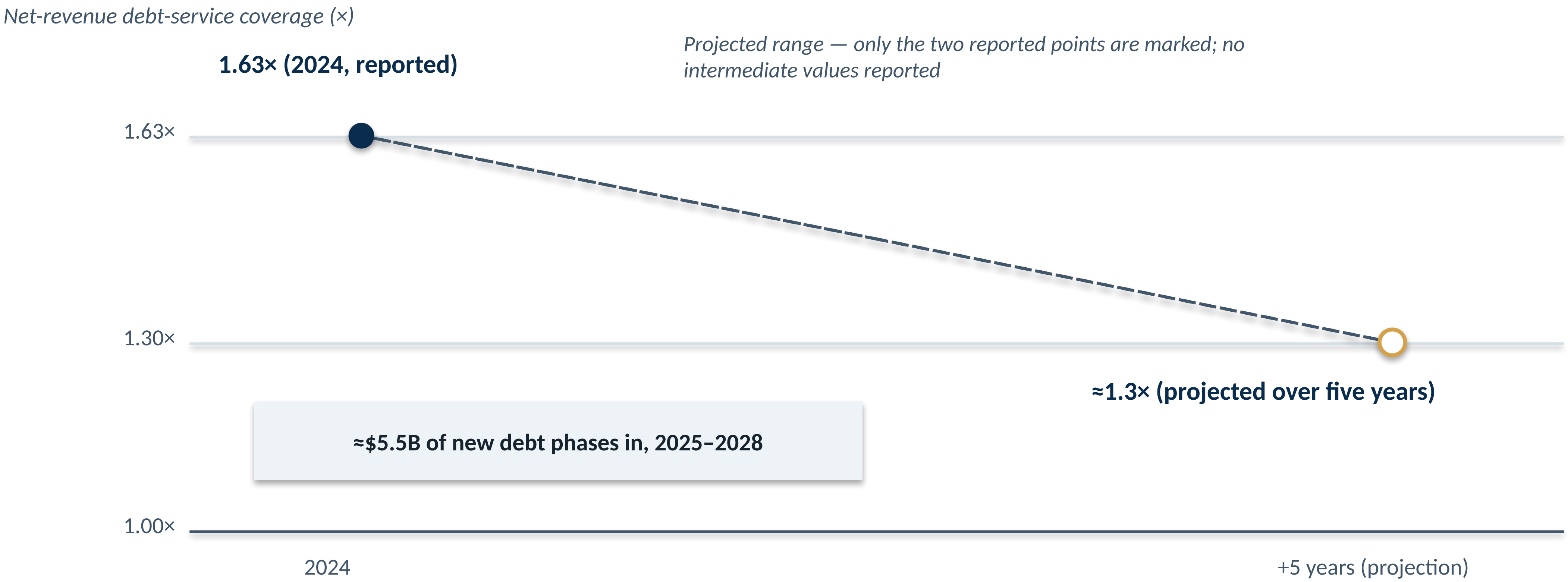
5
Design Manual publication binds consultants and contractors

6
Commissioning under representative traffic, with cure period

First 90 days: commission the Concourse E intelligibility and ambient baseline at the fall 2026 opening. Acceptance floor: STI ≥ 0.45 at ≥ 90% of locations per acoustically distinguishable space, average ≥ 0.50, measured per IEC 60268-16:2020 under representative traffic.

Sources: path per the verified report; Design Manual authority per MWAA (mwaa.com); Concourse E per MWAA project disclosures (flydulles.com), verified Aug 11, 2026. AUL majority-in-interest provisions not verified from primary text — the path's flagged contingency.

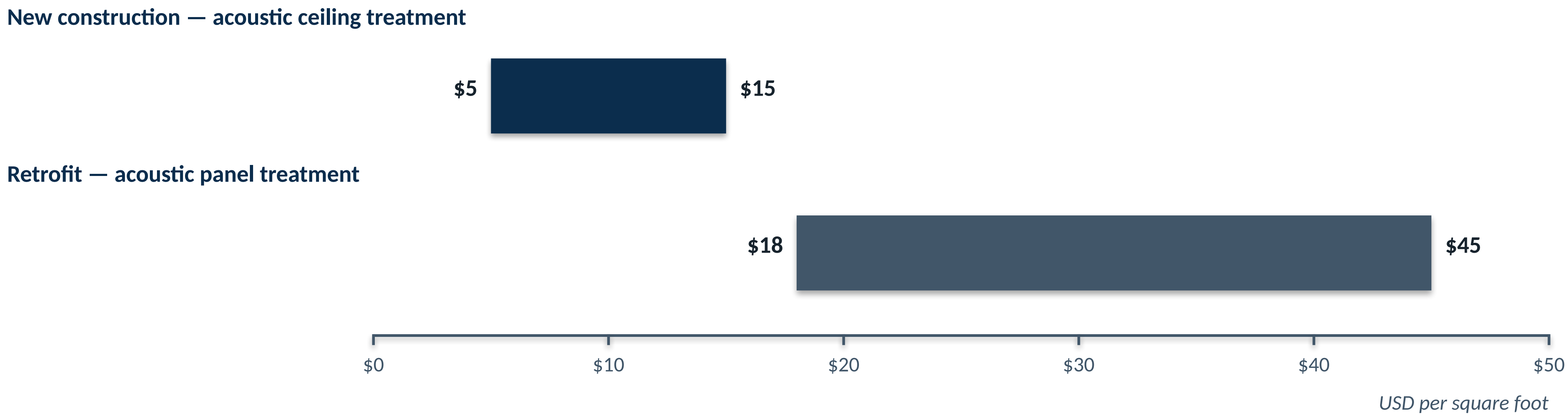
Read-ahead: the coverage window is why the amendment targets work still in programming



Added specification cost inside programs already in design cuts against coverage; specification at programming stage does not.

Sources: Moody's projection as reported by The Bond Buyer, 'MWAA goes to market' (2025; subscription-gated). Underlying rating-agency reports are the primary citation-grade source — confirm before board-facing publication.

Read-ahead: specify at design time — the retrofit premium at commercial benchmarks



Non-airport commercial benchmarks, medium confidence — airside premiums would raise both ranges.

The ranges point one direction: specify at design time.

Sources: commercial pricing guides — designtransitionstudio.com (new-construction acoustic ceiling, 2026 guide); acousticmod.com (retrofit vs. new construction; page since relocated). Non-airport, directional benchmarks.

Read-ahead: one standard, two very different anchor-tenant problems

IAD — INTERNATIONAL-CONNECTION HUB

- 29.01M passengers in 2025; 10.53M international
- Fastest 2025 growth among the 50 largest US airports
- United: ≈67% of passenger traffic; 50 jetbridge-equipped gates in Concourses C and D

DCA — SLOT-CONTROLLED DOMESTIC O&D

- American: ≈53% market share (end-2024)
- 2025 traffic declined — the January collision, softened government-related travel, a weeks-long federal shutdown
- Podium PA effectively airline-owned at preferential-use gates

Airline briefing reported by Cranky Flier (Aug 4, 2026): **fully loaded IAD transformation cost per enplanement ≈$90.64 vs. $12.88 today** — analyst-reported airline positioning, not an MWAA figure.

Sources: Simple Flying (United IAD share and gates, verified Aug 11, 2026; American DCA share, March 2025) — directional trade-press figures; IAD traffic per MWAA reporting via Simple Flying and FFXnow; CPE figures per Cranky Flier (Aug 4, 2026), attributed as analyst commentary.

Read-ahead: half the soundscape is tenant-owned — governance is the technology

HOLDROOM — conceptual, not a floorplan

MWAA-controlled overhead PA zones

Governed by the Design Manual today; the amendment adds intelligibility floors, ADS mapping, and audit-logged governance.

Airline podium PA (preferential-use gate)

Amendment written to reach it — contingent on AUL majority-in-interest verification.

Concessionaire TVs and audio

Lease terms; ambient thresholds set in the amendment.

AUDITABLE GOVERNANCE THE AMENDMENT CAN SPECIFY

- Centralized management
- Replay and audit logs of every broadcast
- Elimination of manual-initiation errors
- Templated multi-language delivery

Enforcement: annual acoustic inspections keyed to lease renewal · cure notices · lease-management escalation for pattern noncompliance.

Sources: airline gate control per trade-press reporting (Simple Flying); AUL structure per MWAA press release, majority-in-interest provisions not verified from primary text; governance platform capabilities per ACI-NA (Feb 27, 2026) — industry advocacy, directional.

Read-ahead: Concourse E carries the first commissioned baseline this fall

Days 1–30
Primary-text verification

Design Manual acoustic and PA content; AUL majority-in-interest provisions (Office of Engineering).

Fall 2026
Concourse E baseline

≈435,000 sq ft, 14 gates. Commission the intelligibility and ambient baseline under representative traffic at opening.

This capital cycle
Amendment issued

Board information item; escalation to resolution only on material airline objection.

Phase-dependent
Reference application

DCA Terminal 1 concourse replacement — subject to phase verification, not verified in this research pass; fallback: earliest IAD C/D element still in programming.

The Concourse E baseline becomes the number every future MWAA facility is measured against.

Sources: Concourse E per MWAA project disclosures (flydulles.com, verified Aug 11, 2026); reference-application phasing per the verified report, flagged unverified; open items listed on the final page of this packet.

Read-ahead: an empty-building reading is not a commissioning result

1 Empty-building STI baseline

Measured per IEC 60268-16:2020 — necessary, but not a commissioning result.

2 Representative-traffic verification

Required because the standard's own scope (§7.13, §8.9.3) does not cover fluctuating background noise — and terminals are the paradigm fluctuating-noise environment.

3 Cure period and re-measurement

Verification-with-cure-period, not an occupancy condition — keeps acceptance off the occupancy critical path.

Related floors: 29 CFR 1910.95 — hearing-conservation program at 85 dB(A) eight-hour TWA · ANSI/ASA S12.60 — 35 dB(A) core-learning-space background, the reference benchmark for MWAA-defined low-stimulation zones only (reference, not terminal ambient).

Sources: IEC 60268-16:2020 scope limitation (standard paywalled; scope confirmed in records citing §7.13, §8.9.3); 29 C.F.R. § 1910.95 (OSHA); ANSI/ASA S12.60 (paywalled; 35 dB(A) figure corroborated).

Read-ahead: 'the floors already exist' is half-right — here is the marginal content

WHAT EXISTING FLOORS REQUIRE

NFPA 72 — emergency voice intelligibility only; no ADS mapping as a general deliverable, no routine-PA governance

2010 ADA Standards — receiver counts, not intelligibility floors for daily announcements

ACRP 175 — guidance, not code

WHAT ONLY THE AMENDMENT ADDS

- ADS map at each design milestone
- Representative-traffic commissioning
- Audit-logged governance of routine PA
- Channel-redundancy floors
- IROPS override
- Tenant-audio boundaries

The serious objection: premature specification before program-brief maturity drives change orders, and hard commissioning gates on the critical path in this coverage window would be a bond-rating conversation. Answered by cure-period commissioning and milestone — not occupancy — deliverables.

Sources: counter-case and rebuttal per the verified report; regulatory scope per NFPA 72, 2010 ADA Standards, and ACRP Research Report 175 as corroborated in the evidence base.

Read-ahead: what this research pass did not verify

Open verification item	Owner	Closure route
Current Design Manual acoustic, PA-zoning, and STI content	Office of Engineering	Read Volume 2 and applicable specification sections (public, mwaa.com)
2025 AUL majority-in-interest thresholds and cost-recovery categorization	Office of Engineering with airline relations	Primary agreement text
DCA Terminal 1 concourse replacement project phase	Capital program office	Capital program status reporting
SFO Quiet Airport measurement methodology	Research follow-up	SFO Airport Commission minutes or program report
Rating-agency primary reports behind the Bond Buyer coverage figures	Finance	Moody's, Fitch, and S&P MWAA reports

Every figure in this packet carries a documented source; the full source appendix is in the companion report. Nothing on this page is asserted as verified.

Sources: open items per the report's evidence review. Closing them is the first 90 days of assigned work.